DriveSim: Drive-Thru Simulation

CSE 10L MODELING AND SIMULATION

Introduction to Problem Definition

DRIVE-THRU SERVICE CHALLENGES

- Service imbalance due to fluctuating arrivals
- Fixed capacity leads to long queues
- Order-window bottlenecks cause congestion
- Uneven staff utilization impacts efficiency
- Goal: Test demand and staffing changes

Simulation Objectives for Performance Optimization

KEY GOALS OF DRIVE-THRU SIMULATION ANALYSIS

- Measure average queue waiting time effectively
- Identify bottlenecks in service workflow
- Evaluate service-window utilization and throughput rates
- Compare peak demand against added capacity scenarios
- Recommend changes to reduce wait times efficiently

System Model Design

DRIVE-THRU SIMULATION

The model encompasses:

- Vehicle Arrival: FIFO Queue
- Key Events: Arrival, Order, Payment, Pickup, Departure
- Resources: Order Station, Payment Window, Pickup Window
- State Variables: Vehicles in System, Queue Length, Busy Time, Completed Vehicles

Assumptions and Input Data

KEY PARAMETERS OVERVIEW

- Arrivals follow a Poisson-style process
- Normal arrival rate: 12 vehicles/hour
- Peak demand from 11:00 AM to 2:00 PM
- Service times: Order (1-3 min), Payment (0.5-1 min), Pickup (1-2 min)
- Baseline: 1 order station, 1 payment window

Implementation Details of DriveSim

UTILIZING SIMPY AND TKINTER

- Built with the **SimPy** discrete event simulation framework
- Interactive **Tkinter** desktop interface for user control
- Adjustable parameters such as arrival rates and lane capacity
- Animated view shows vehicle movements through the system
- Outputs include event logs and analysis summaries

Experimental Results Overview

BASELINE PERFORMANCE

The baseline scenario served 297 vehicles with an average wait time of **2.02 minutes**, highlighting initial service effectiveness.

PEAK DEMAND INSIGHTS

During peak demand, **344 vehicles** were served, but average wait times increased to **2.86 minutes**, indicating significant congestion issues.

OPTIMIZED CAPACITY OUTCOMES

With optimized capacity, **345 vehicles** were served without rejections, achieving an impressive average wait time of only **0.29 minutes**.

Key Findings on Queue Wait Times

COMPARATIVE ANALYSIS

The average queue wait increased by **42 percent** during peak demand, highlighting significant service challenges faced by the drive-thru system.

OPTIMIZATION RESULTS

The optimized scenario achieved a **90 percent reduction** in wait time, demonstrating the effectiveness of adding capacity to enhance service efficiency.

Conclusions

INSIGHTS AND RECOMMENDATIONS

Optimizing the drive-thru system is essential; enhancing order capacity significantly reduces wait times and improves customer satisfaction, thereby increasing overall efficiency.

Thank You!